

Realtime Keylogger & Bruteforce Sentinel

Comprehensive Project Report

Abstract

Organizations face increasing threats from endpoint-based malware and unauthorized authentication attempts. Keyloggers are capable of capturing sensitive user information, while brute-force attacks target authentication mechanisms to gain unauthorized access.

Realtime Keylogger & Bruteforce Sentinel was developed as a Security Operations Center (SOC) monitoring framework capable of detecting both attack categories in real time using Sysmon, Winlogbeat, Elasticsearch, and Kibana Security.

The framework provides centralized log collection, event correlation, custom detection rules, automated alerting, and incident investigation capabilities.

Introduction

Security analysts require visibility into endpoint activities to identify malicious behavior before it leads to compromise.

Traditional operating system logs often provide insufficient visibility into:

- Process creation
- Command-line execution
- Authentication abuse
- Malware activity

To improve detection capabilities, this project integrates endpoint telemetry with SIEM-based monitoring and alerting.

The framework focuses on detecting:

- Keylogger Activity
- Brute Force Attempts
- Suspicious Processes
- Authentication Anomalies

Project Objectives

The primary objectives of this project are:

- Monitor Windows endpoint activity
- Collect detailed Sysmon telemetry

- Forward security logs to Elasticsearch
- Detect keylogger behavior
- Detect brute-force login attempts
- Generate real-time alerts
- Deliver email notifications
- Support SOC investigations
- Improve endpoint visibility

Technology Stack

Endpoint Monitoring

Sysmon

Log Forwarding

Winlogbeat

SIEM Platform

Elasticsearch

Kibana Security

Alerting

Kibana Detection Rules

Email Notifications

Security Concepts

Detection Engineering

Threat Hunting

Incident Response

SOC Monitoring

System Architecture

Windows Endpoint

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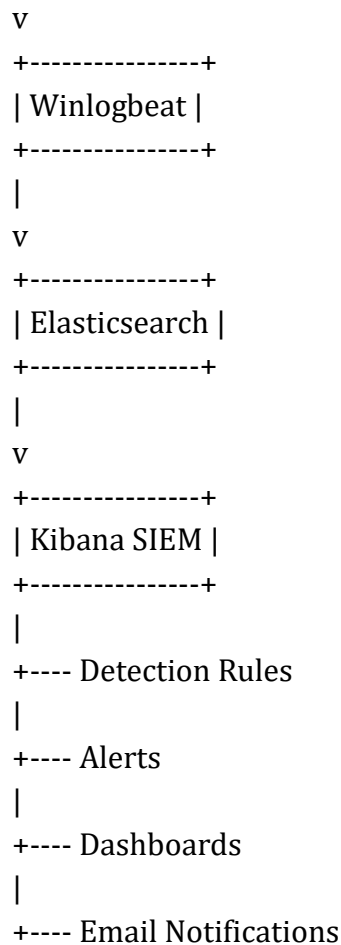
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Phase 1: Sysmon Deployment

Sysmon was deployed to enhance Windows logging capabilities.

Purpose:

- Process Creation Monitoring
- File Activity Monitoring
- Command-Line Monitoring
- Security Event Collection

Benefits:

- Enhanced endpoint visibility
- Detailed attack telemetry
- Better detection opportunities

Sysmon Configuration

A customized Sysmon configuration was deployed.

Monitored Events Included:

Event ID 1

Process Creation

Event ID 3

Network Connection

Event ID 7

Image Loaded

Event ID 11

File Creation

Event ID 13

Registry Modification

Event ID 22

DNS Query

These events provide comprehensive visibility into endpoint activity.

Phase 2: Log Collection

Winlogbeat was configured to collect:

- Windows Security Logs
- Sysmon Logs
- Event Logs

Collected events were forwarded to Elasticsearch for centralized analysis.

Benefits:

- Centralized logging
- Real-time processing
- Searchable security events

Phase 3: Elasticsearch Integration

Elasticsearch served as the centralized data repository.

Functions:

- Log Storage
- Event Indexing
- Fast Search

- Historical Analysis

The environment successfully ingested over 570,000 security events during testing.

Phase 4: Kibana Security Configuration

Kibana Security was configured as the primary SOC interface.

Capabilities:

- Alert Management
- Threat Investigation
- Security Dashboards
- Detection Rules
- Event Correlation

Phase 5: Keylogger Detection Engineering

A custom detection rule was created to identify keylogger activity.

Detection Logic:

- Suspicious process execution
- Known keylogger indicators
- Malicious monitoring software
- Abnormal endpoint behavior

Alert Characteristics:

Severity: High

Risk Score: 80

Result:

Real-time alerts generated when suspicious activity was detected.

Phase 6: Brute Force Detection Engineering

A custom brute-force detection rule was implemented.

Detection Logic:

- Multiple failed logins
- Repeated authentication failures
- Login abuse patterns
- User targeting behavior

MITRE ATT&CK Technique:

T1110 – Brute Force

Alert Characteristics:

Severity: High

Risk Score: 80

Phase 7: Alert Generation

When detection criteria were met:

1. Alert created
2. Event correlated
3. Severity assigned
4. Incident logged
5. Analyst notified

Observed Alerts:

- Keylogger
- Brute Force / Multiple Failed Logins

Phase 8: Email Notification System

The framework was configured to automatically notify analysts.

Alert Email Contents:

- Alert Name
- Event Information
- Timestamp
- Process Details
- Host Information
- Investigation Notes

Benefits:

- Faster response time
- Improved visibility
- Automated incident notification

Security Dashboards

Custom Kibana dashboards were created to visualize:

Alert Distribution

Displays:

- Alert Categories
- Alert Counts
- Severity Levels

Threat Trends

Displays:

- Historical Activity
- Detection Frequency
- Investigation Metrics

Top Alert Sources

Displays:

- Host Systems
- User Activity
- Event Volume

Threat Hunting Capabilities

Analysts can investigate:

- Process Execution Chains
- User Activity
- Authentication Events
- Security Alerts
- Endpoint Telemetry

Skills Demonstrated

- Detection Engineering
- Elastic SIEM
- Kibana Security
- Elasticsearch Administration
- Sysmon Configuration

- Winlogbeat Deployment
- Threat Hunting
- Incident Response
- Alert Engineering
- Security Analytics
- Endpoint Monitoring
- SOC Operations

MITRE ATT&CK Mapping

Keylogger Detection

T1056 – Input Capture

Brute Force Detection

T1110 – Brute Force

Process Monitoring

T1059 – Command and Scripting Interpreter

Endpoint Visibility

T1082 – System Information Discovery

Learning Outcomes

Through this project, the following competencies were developed:

- SIEM Deployment
- Endpoint Security Monitoring
- Log Management
- Detection Rule Creation
- Alert Engineering
- Security Investigation
- Threat Hunting
- Incident Response Workflows
- SOC Operations

Conclusion

Realtime Keylogger & Bruteforce Sentinel successfully demonstrates the design and implementation of a real-time threat detection framework capable of monitoring Windows endpoints, detecting suspicious activities, generating alerts, and supporting incident investigations.

By integrating Sysmon, Winlogbeat, Elasticsearch, and Kibana Security, the project provides enterprise-style visibility into endpoint behavior and showcases practical skills in SIEM engineering, detection engineering, threat hunting, and SOC operations.

The project reflects real-world Security Operations Center workflows and demonstrates capabilities directly relevant to SOC Analyst, Detection Engineer, and Security Analyst roles.

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